

MACHINE LEARNING CAPSTONE PROJECT

Model Comparison & Explainability

Medical Insurance Cost Classification

Final Project Report

1. Executive Summary

This project develops an end-to-end machine learning classification solution using the Medical Cost Personal Dataset from a previous internship task. The original dataset is a regression dataset containing individual characteristics and medical insurance charges. For this capstone, the problem was adapted into binary classification: High Cost versus Low Cost.

The median medical charge, 9382.033, was used as the threshold. Five models were evaluated on the same stratified train-test split: Logistic Regression, Random Forest, Decision Tree, XGBoost, and a Stacked Ensemble. XGBoost was selected as the final model with Precision 0.9837, Recall 0.9030, F1-score 0.9416, and ROC-AUC 0.9505. SHAP and LIME were used for explainability, and the model was prepared for Streamlit and Docker deployment.

2. Project Objective

The objective is to demonstrate a complete machine learning workflow from data inspection and preparation through model comparison, final-model selection, explainability, and deployment.

Because the source dataset is originally intended for regression, this report explicitly treats the work as a classification adaptation rather than claiming it is an insurance-claims-fraud dataset.

3. Dataset Description

The dataset contains 1,338 records and seven original columns: age, sex, BMI, children, smoker, region, and charges. The first six variables are model inputs. Charges is used only to construct the target and is excluded from the model to prevent target leakage.

4. Data Exploration and Cleaning

Initial inspection covered dimensions, data types, missing values, duplicates, descriptive statistics, distributions, categorical frequencies, correlations, and medical charges by smoking status. The dataset contained no missing values or duplicate records requiring removal.

5. Target Engineering

The binary target High_Cost was created using the median charge of 9382.033. Values at or above the threshold are labeled 1 (High Cost); values below it are labeled 0 (Low Cost). Charges is then removed from the feature set.

6. Feature Engineering and Preprocessing

Numerical features age, BMI, and children were standardized with StandardScaler. Categorical features sex, smoker, and region were encoded with OneHotEncoder(handle_unknown='ignore'). An 80/20 stratified train-test split with random_state=42 was used consistently for all models.

7. Model Comparison

Five classification approaches were trained and evaluated using Precision, Recall, F1-score, and ROC-AUC. The results are shown below.

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8849	0.9179	0.9011	0.9425
Random Forest	0.9758	0.9030	0.9380	0.9490
Decision Tree	0.8806	0.8806	0.8806	0.8806
XGBoost	0.9837	0.9030	0.9416	0.9505
Stacked Ensemble	0.9606	0.9104	0.9349	0.9466

8. Final Model Selection

XGBoost was selected because it achieved the highest Precision, F1-score, and ROC-AUC. Its Recall was slightly lower than Logistic Regression and the Stacked Ensemble, representing a trade-off between maximizing detection and minimizing incorrect High Cost predictions. The final XGBoost configuration used 200 estimators, max_depth=4, learning_rate=0.05, and random_state=42.

9. Error Analysis

A confusion matrix was used to distinguish false positives from false negatives. This provides a more meaningful assessment than accuracy alone because the two error types may have different consequences.

10. SHAP Explainability

SHAP was used for global and local explanations. The global summary identifies influential transformed features across the test set, while local explanations show how feature values contributed to individual predictions. Three local cases were examined.

11. LIME Explainability

LIME was used as a second local explainability method. Three individual cases were explained to provide complementary local interpretations of the model.

12. Deployment

The final preprocessing pipeline and XGBoost model were saved as final_model.pkl. A Streamlit application accepts the model's input features and returns a High Cost/Low Cost prediction, probability, and SHAP-based explanation. Docker packaging was prepared using a Dockerfile and pinned Python dependencies.

13. Project Deliverables

- Documented Jupyter notebook
- Five-model comparison table
- Saved final XGBoost model
- Global and local SHAP explanations
- Three LIME explanations
- Streamlit prediction and explainability app
- Dockerized application
- Model comparison and explainability outputs

14. Strengths and Limitations

Strengths

- Consistent train-test evaluation across five models.
- Multiple performance metrics.
- SHAP and LIME explainability.
- Saved preprocessing pipeline and model.
- Streamlit and Docker deployment.

Limitations

- Dataset size is 1,338 records.
- High Cost is an artificial median-based target.
- The threshold is not an industry or clinical standard.
- Additional real-world/domain validation would be required before practical decision-making.

15. Conclusion

This project demonstrates the complete path from raw medical insurance data to an explainable and deployable classification system. XGBoost provided the strongest overall performance among the five evaluated approaches, while SHAP and LIME made individual and global model behavior more interpretable.